Title of post: Bioinformatics database developer

Job Purpose: To co-ordinate and manage the development of BGCI's PlantSearch database and associated databases

Duration: 1 year subcontracted Fixed Term Appointment (with the potential to extend)

Remuneration: Tenders invited

Job Description

There are an estimated 2500 botanic gardens and arboreta in the world attracting 750 million visitors each year. Botanic Gardens Conservation International (BGCI) is at the centre of this network. Our members include the largest, most renowned gardens on the planet – Kew, New York, Missouri, Singapore, Sydney and Shanghai –and they also include many smaller gardens situated in the world's plant diversity hotspots. All of these member gardens share a commitment to making sure that no plant species becomes extinct and they have a combined workforce of many thousands of horticulturalists and scientists working towards that end.

BGCI provides support to its members in many different ways: we are an **advocate** for the botanic garden community, promoting the role of botanic gardens in plant conservation policy, practice and education; we **connect people**, co-ordinating efforts and bringing people together to exchange ideas; we are a **knowledge hub**, providing training, learning resources and professional support; and we provide **funding** and **technical support** to efforts aimed at saving plants.

Central to BGCI's knowledge hub function is its databases: GardenSearch, the only global directory of botanical institutions (https://tools.bgci.org/garden_search.php); PlantSearch, the only global register of the plants grown in botanic gardens (https://tools.bgci.org/plant_search.php); GlobalTreeSearch, the only global list of all tree species and their country level distributions (https://tools.bgci.org/global_tree_search.php) and ThreatSearch, the only global database of all known threat assessments for plants (https://tools.bgci.org/global_tree_search.php).

The primary purpose of this role is to oversee improvements in BGCI's PlantSearch database, its integration with BGCI's other databases and the improvement of BGCI's web-based database user experience. We are looking for a Bioinformatics database developer to lead this process. Specific tasks will include:

1. **Leading the development of BGCI's PlantSearch database, including integration with BGCI's other databases.** This will include**:**
    - Co-ordinating outsourced database development, working closely with developers, BGCI project managers and budget holders.
    - Managing the development of an accessions level module for responsible and informed exchange of plant material for conservation and research.
    - Managing the development of PlantSearch 2, which incorporates a range of tools and improvements to the current PlantSearch database, including incorporation of a taxonomic standard, improved data contribution and request capabilities, a new climate resilience tool, a new propagations protocol tool, and integration and alignment with priority data including threat status and tree status.
    - Integration of elements of BGCI's other databases into PlantSearch.
    - Evaluating and testing new products and software applications under development.

2. **Leading the development of BGCI's web-based database user interface and improvements in user experience.** This will include**:**
    - Consulting with BGCI's database users
    - Developing and testing database user queries and entry points to optimize the user experience
    - Overseeing the improvement of BGCI's database web interfaces, including in-house and outsourced web design and programming.
    - Working closely with BGCI's technical teams, project managers and budget holders.

Person Specification

BGCI is seeking to appoint a person with bioinformatics, database and website design skills as well as excellent communications and networking skills. Candidates for this post will require a proven track record working in bioinformatics for a conservation or scientific organisation, preferably in plant conservation. The post holder will demonstrate sensitivity to complex, multicultural and multinational institutions and structures.

Education and Experience

Essential

- Bachelor's degree in Computer Science, Information Systems, or equivalent.
- 3+ years relevant work experience in bioinformatics.
- Bioinformatics project management experience preferred with experience in improving processes, systems integration, and adoption of technology platforms to achieve successful and sustainable outcomes.
- Demonstrated experience with troubleshooting principles, methodologies, and issue resolution techniques.

Desirable

- Experience of working for a not for profit organisation

- Knowledge of botanic gardens, nature conservation, zoos, museums or related disciplines ideally gained through direct personal experience

Skills and competences

Essential

- Technical experience with designing, building, installing, configuring, and supporting biological databases, including MySQL servers. Advanced SQL administration skills strongly preferred.
- Website development expertise, including bioinformatics user interfaces, mobile applications, web services and web sites.
- Versatile with a combination of database and systems skills.
- Highly self-motivated, directed, with keen attention to detail.
- Ability to work with BGCI staff to interpret needs and goals for technical implementation.
- Ability to develop, retain, and interpret technical documentation for training and end-user procedures.
- Ability to investigate and solve a wide range of computing issues as required.
- Ability to effectively prioritize and execute tasks in a high-pressure environment.
- Customer service-oriented individual with excellent oral and written communication skills, and the ability to effectively work in a team-oriented, collaborative environment.
- Knowledge of trends in technology relating to database and software applications.
- Knowledge of applicable data privacy practices and laws.

Desirable

- Front end and back end web development experience including but not limited to expertise in HTML/CSS, JavaScript/JQuery, version control/GIT, Laravel, Python, Node.js, PHP, and SQL.
- Experience in different Content Management Systems including but not limited to WordPress.
- Command of additional languages.
- Knowledge of plant conservation approaches and methodologies
- Knowledge of plant nomenclature and taxonomy strongly desired